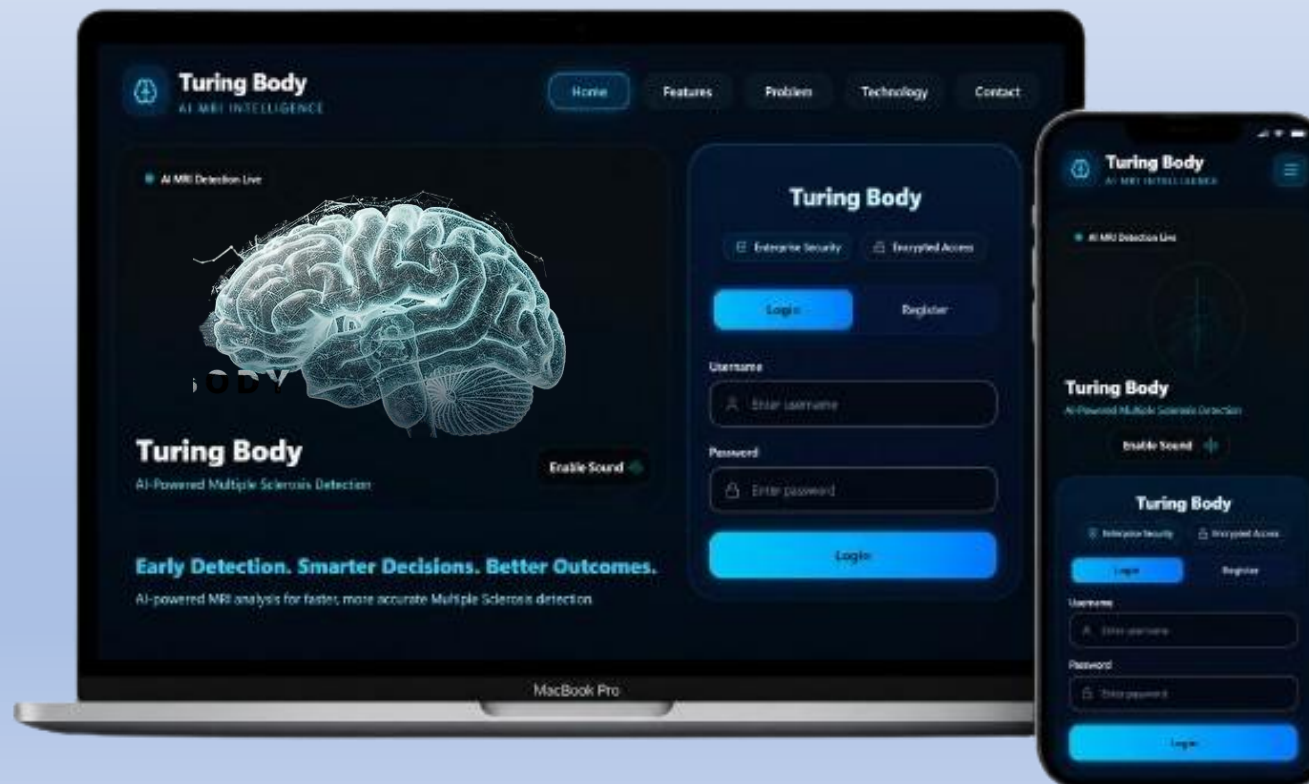


# Turing Body

## AI-Based Early Detection of Multiple Sclerosis Using MRI Image Analysis



# Introduction of Turing Body

❑ Turing Body is an AI-powered medical imaging platform developed for the automated detection and segmentation of Multiple Sclerosis (MS) lesions from MRI brain scans.

The system combines:

- Deep Learning
- Medical Image Processing
- Spring Boot Backend APIs
- React Frontend Framework

❑ The core intelligence of the system is based on the U-Net Deep Learning architecture, which performs pixel-level lesion segmentation from MRI FLAIR images. The platform allows doctors to upload MRI scans, automatically analyze lesion regions, estimate disease severity, visualize segmentation results, and generate PDF-based medical reports.

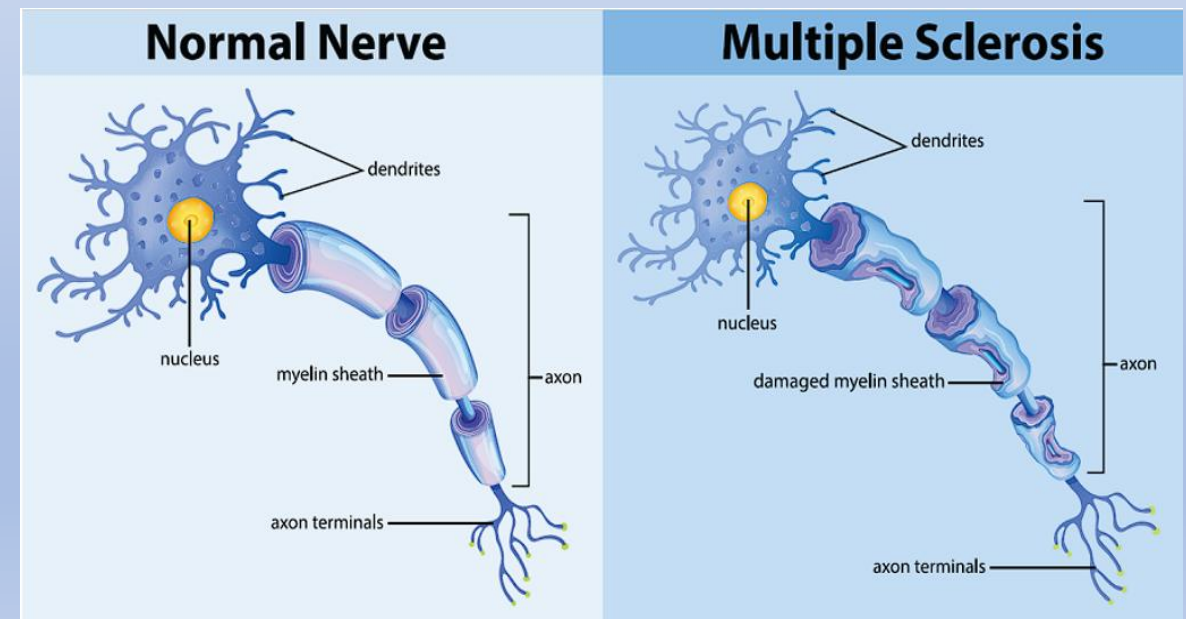
Turing Body was designed to improve:

- lesion analysis efficiency,
- segmentation automation,
- medical visualization,
- and AI-assisted diagnosis workflows.

# Problem Statement

- Multiple Sclerosis (MS) is a chronic neurological disorder that affects the central nervous system by damaging the myelin sheath — the protective covering surrounding nerve fibers in the brain and spinal cord.
- MRI (Magnetic Resonance Imaging) is one of the primary diagnostic tools used to identify MS lesions. However, manual MRI lesion analysis presents several challenges:
  - Time-consuming examination process
  - Difficulty in identifying small and irregular lesions
  - Dependency on radiologist expertise
  - Risk of human error and inconsistent interpretation
  - Challenging analysis for large MRI datasets

Therefore, there is a need for an intelligent automated system that can accurately detect, segment, and analyze MS lesions from MRI brain scans using Artificial Intelligence and Deep Learning techniques.



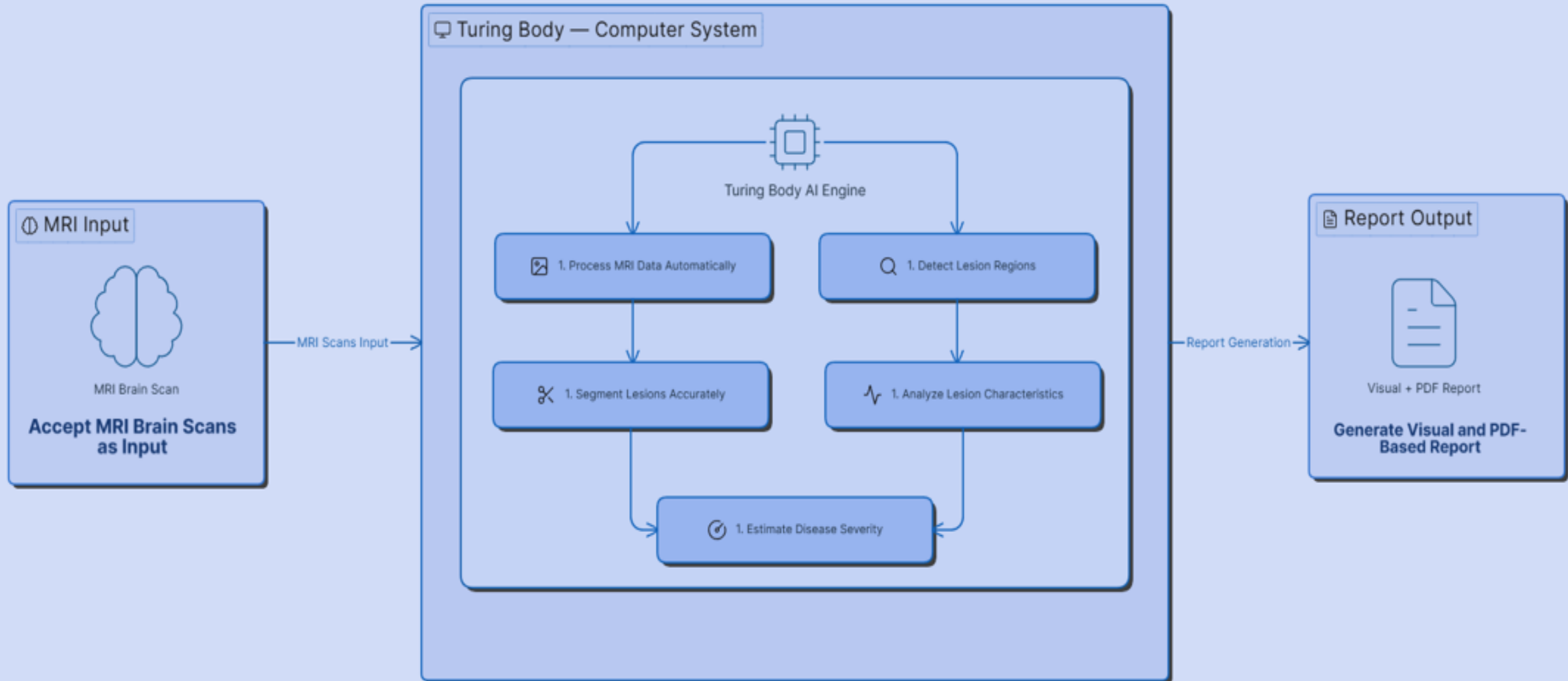
# Existing System Limitations

| Traditional MRI Analysis                                                                                                                                                                                                                             | Traditional Machine Learning                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>▪ Requires manual MRI inspection</li><li>▪ Time-consuming process</li><li>▪ Depends on radiologist expertise</li><li>▪ Inconsistent lesion interpretation</li><li>▪ Difficult for large MRI datasets</li></ul> | <ul style="list-style-type: none"><li>▪ Requires manual feature extraction</li><li>▪ Poor pixel-wise segmentation</li><li>▪ Limited lesion localization</li><li>▪ Lower segmentation accuracy</li><li>▪ Cannot detect fine lesion boundaries</li></ul> |

## Result

Traditional approaches fail to provide fast, accurate, and automated MRI lesion segmentation, making MS diagnosis more challenging and time-intensive.

# Objective of Turing Body



## Dataset Used

- The project utilizes MRI FLAIR brain scans along with corresponding ground truth lesion masks for supervised Deep Learning-based lesion segmentation.

- Dataset Characteristics
- Dataset Type: MRI FLAIR Brain Scans
- Ground Truth: Expert-annotated lesion masks
- File Format: .nii (NIfTI Medical Imaging Format)

## Dataset Contains

- Original MRI brain scan volumes
- Expert-labeled lesion segmentation masks
- The annotated lesion masks serve as ground truth labels for training the U-Net segmentation model.

## Data Preprocessing Pipeline

- To prepare MRI data for Deep Learning segmentation, a preprocessing pipeline was implemented to standardize and optimize the input data.

### Preprocessing Steps:

- MRI File Reading
- Slice Extraction from 3D MRI Volumes
- Intensity Normalization
- Image Resizing to  $128 \times 128$
- Lesion Mask Preparation
- Train-Test Dataset Splitting

# Core Model U-Net Architecture

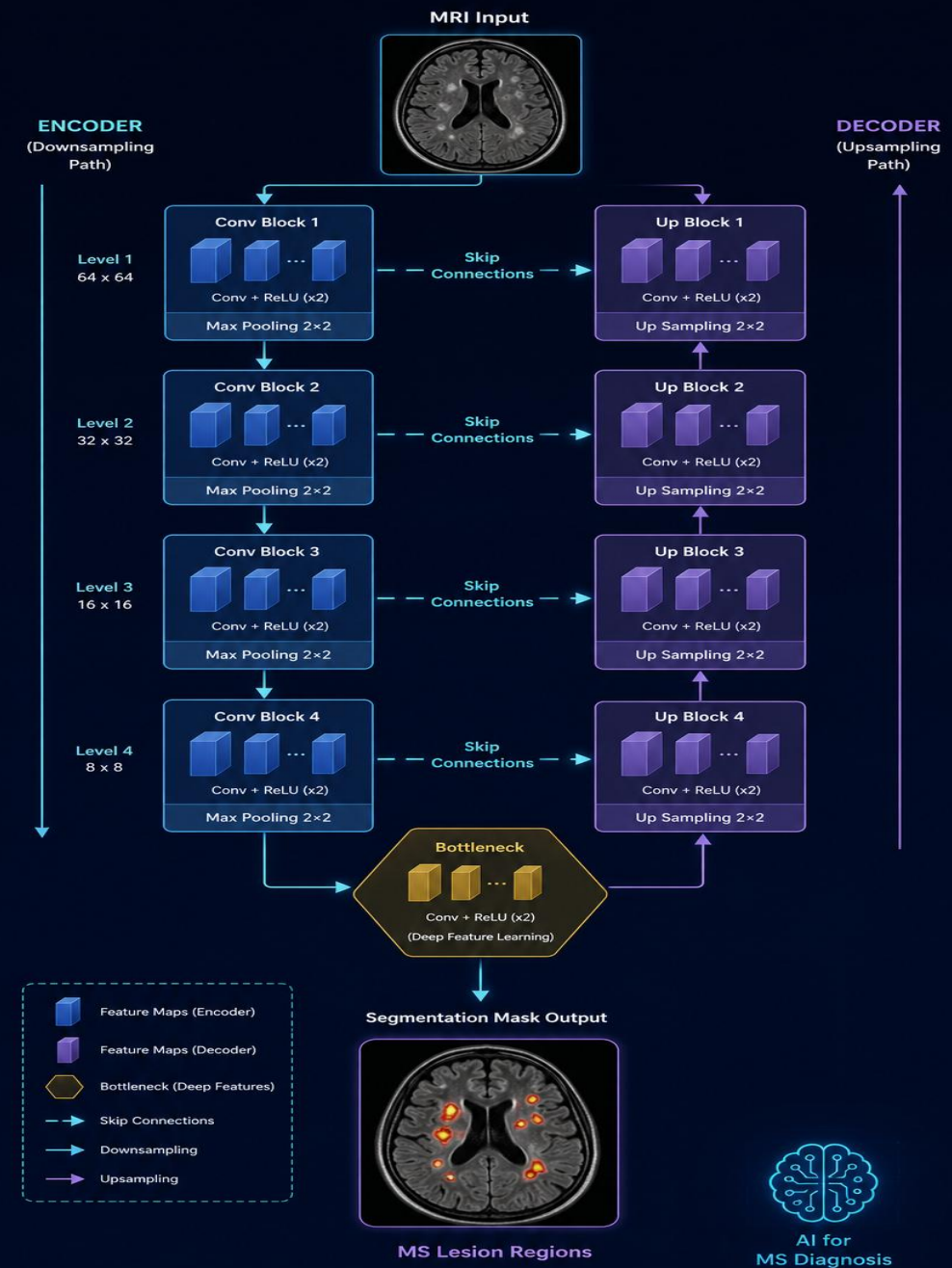
## U-Net Convolutional Neural Network

- The proposed system uses a **U-Net Deep Learning architecture** for automated Multiple Sclerosis lesion segmentation from MRI brain scans.
- U-Net is a specialized Convolutional Neural Network (CNN) designed specifically for **biomedical image segmentation**, where precise pixel-level localization is required.

### Why U-Net Was Selected

- Specifically designed for medical image segmentation
- Performs effectively on small medical datasets
- Provides accurate lesion localization
- Uses skip connections for fine feature recovery
- Produces pixel-level segmentation predictions
- Widely used in biomedical imaging research
- Preserves important spatial information

## U-Net Architecture for Multiple Sclerosis MRI Lesion Segmentation



# Model Performance & Metrics



## EVALUATION METRICS for Lesion Segmentation

### WHAT IS BEING EVALUATED?

Predicted  
Lesion Mask

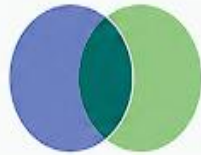
Ground Truth  
Lesion Mask



VS

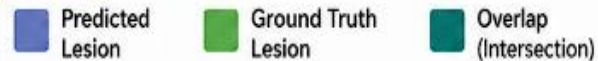


### OVERLAP



Higher overlap  
= **Better** segmentation  
accuracy

### LEGEND



These metrics help **quantify** how well the predicted lesions match the ground truth, guiding the model to learn **better** and segment more accurately.



**COMBINATION OF THESE METRICS** leads to accurate, reliable, and robust lesion segmentation models.

## RIGHT SIDE — EVALUATION METRICS

1



### DICE COEFFICIENT

- Measures overlap between:
  - predicted** lesion mask
  - ground truth** mask
- Higher Dice = **Better** segmentation accuracy

2



### DICE LOSS

- Minimizes segmentation error
- Improves lesion overlap quality
- Helps the model learn better boundaries and shapes

3



### BINARY CROSS ENTROPY (BCE)

- Measures pixel-wise classification error between:
  - lesion pixels**
  - non-lesion pixels**
- Improves segmentation **precision**



### Validation Performance

Validation Dice Score

**56%**



Measures segmentation  
quality on unseen MRI  
scans.



### Training Performance

Training Dice Score

**71%**



Indicates effective  
lesion feature  
learning.



## Performance Analysis



Stable lesion segmentation on unseen MRI scans



Effective feature learning using U-Net



Reasonable generalization for limited medical datasets



Improved lesion localization through Deep Learning



# TECHNOLOGY STACK – TURING BODY

AI-Powered MRI Lesion Segmentation | Secure Backend | Modern Web Interface | Scalable Deployment

| MACHINE LEARNING & AI                                                              |                    |                         |
|------------------------------------------------------------------------------------|--------------------|-------------------------|
| Technology                                                                         |                    | Purpose                 |
|    | Python             | ML Development          |
|    | TensorFlow / Keras | Deep Learning Framework |
|    | U-Net              | MRI Lesion Segmentation |
|    | OpenCV             | Image Processing        |
|    | NumPy              | Numerical Computation   |
|    | SciPy              | Lesion Analysis         |
|  | nibabel            | MRI .nii File Handling  |

| BACKEND TECHNOLOGIES                                                                |                 |                                |
|-------------------------------------------------------------------------------------|-----------------|--------------------------------|
| Technology                                                                          |                 | Purpose                        |
|    | Spring Boot     | Backend Framework              |
|    | Spring Security | Authentication & Authorization |
|    | JWT             | Secure Token Authentication    |
|    | Spring Data JPA | Database ORM                   |
|    | MongoDB         | NoSQL Database                 |
|    | Maven           | Dependency Management          |
|  | Docker          | Containerization               |

| FRONTEND TECHNOLOGIES                                                               |                         |                     |
|-------------------------------------------------------------------------------------|-------------------------|---------------------|
| Technology                                                                          |                         | Purpose             |
|  | Next.js                 | Frontend Framework  |
|  | React.js                | UI Development      |
|  | Tailwind CSS            | Responsive Styling  |
|  | Axios / Fetch API       | API Communication   |
|  | TypeScript / JavaScript | Frontend Logic      |
|  | Vercel                  | Frontend Deployment |

| DEPLOYMENT & TOOLS                                                                  |              |                         |
|-------------------------------------------------------------------------------------|--------------|-------------------------|
| Tool / Platform                                                                     |              | Purpose                 |
|  | Git          | Version Control         |
|  | Google Colab | Model Training          |
|  | VS Code      | Development Environment |
|  | Postman      | API Testing             |
|  | Render       | Backend Deployment      |
|  | Railway      | Backend Deployment      |



## POWERING AN END-TO-END AI SYSTEM



AI models segment lesions from MRI scans



Secure backend handles data & users

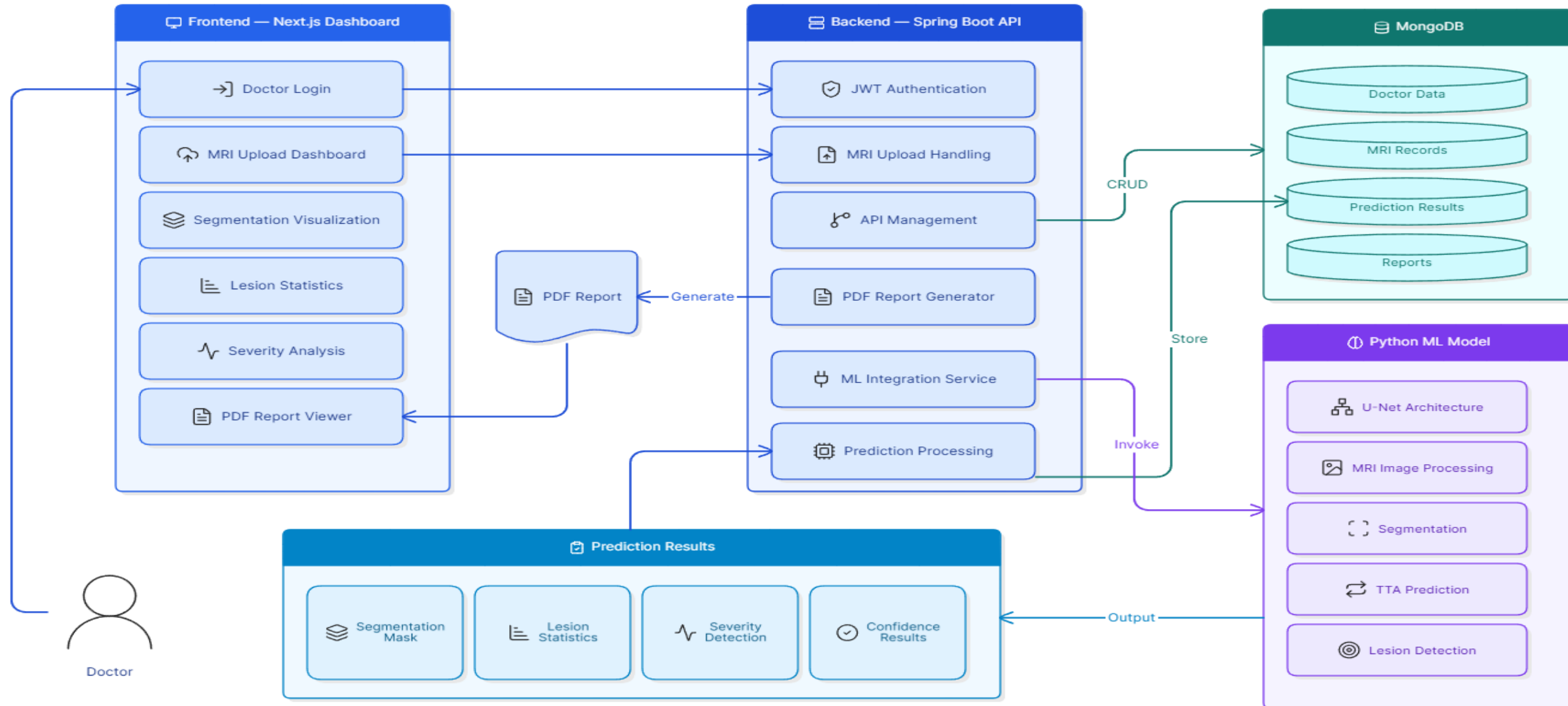


Modern frontend delivers seamless UX



Deployed & scalable for real-world use

# System Architecture of Turing Body



# Background Study(Research Papers studied)

| Sr. number | Paper Title                                                                   | Publisher | Year | Take-away points                                                         |
|------------|-------------------------------------------------------------------------------|-----------|------|--------------------------------------------------------------------------|
| 1          | Deep Learning-Based Multiple Sclerosis Lesion Segmentation in MRI             | IEEE      | 2021 | CNN models improve detection of MS lesions in brain MRI scans            |
| 2          | Automated Detection of Multiple Sclerosis Using Convolutional Neural Networks | Springer  | 2020 | Deep learning provides higher accuracy than traditional ML methods       |
| 3          | MRI Brain Image Classification for Neurological Disorders Using AI            | Elsevier  | 2022 | AI enables early and consistent diagnosis of neurological diseases       |
| 4          | Medical Image Analysis Using Deep Learning Techniques                         | IEEE      | 2023 | Deep learning enhances feature extraction and classification performance |

# Existing Research vs Proposed System

| System                                      | Accuracy / Performance                      | Limitations                                                              |
|---------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------|
| Traditional Manual MRI Analysis             | Depends on radiologist expertise            | Time-consuming and inconsistent                                          |
| Classical Machine Learning Approaches       | Moderate classification accuracy            | Requires manual feature extraction                                       |
| Existing CNN-Based Systems (2022–2023)      | ~70–90% classification accuracy             | Mostly classification only                                               |
| Medical Image Analysis using Deep Learning  | ~55.93% Dice Segmentation Accuracy          | Limited lesion boundary localization                                     |
| <b>Proposed System – Turing Body (2025)</b> | Automated semantic segmentation using U-Net | Improved lesion localization, severity analysis, and automated reporting |

# References

|                                                                                                                                                               |                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| ➤ O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” Proceedings of MICCAI, pp. 234–241, 2015.       | <a href="#"><u>Unet research paper</u></a>                                          |
| ➤ K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” Proceedings of CVPR, pp. 770–778, 2016.                                | <a href="#"><u>ResNet Research Paper</u></a>                                        |
| ➤ I. Goodfellow, Y. Bengio, and A. Courville, <i>Deep Learning</i> , MIT Press, 2016.                                                                         | <a href="#"><u>Deep Learning Book</u></a>                                           |
| ➤ D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” Proceedings of ICLR, 2015.                                                            | <a href="#"><u>Adam Optimizer Paper</u></a>                                         |
| ➤ F. Chollet, <i>Deep Learning with Python</i> , Manning Publications, 2018.                                                                                  | <a href="#"><u>Deep Learning with Python</u></a><br><br><a href="#"><u>Book</u></a> |
| ➤ A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet Classification with Deep Convolutional Neural Networks,” Proceedings of NIPS, pp. 1097–1105, 2012. | <a href="#"><u>AlexNet Research Paper</u></a>                                       |

# Project View



Live project Link : <https://ai-ms-detection-system.vercel.app/>

Turing Body

AI MRI INTELLIGENCE

Home

Features

Problem

Technology

Contact

AI MRI Detection Live

Turing Body

AI-Powered Multiple Sclerosis Detection

Enable Sound

Enterprise Security

Encrypted Access

Login

Register

Username

Enter username

Password

Enter password

Login

TURING BODY

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

AI Prediction Results

MRI Analysis & Clinical Diagnostic Report

Patient ID: 6a07dd68a0edb90658925caa

Send Report

PDF Report

MRI Scan Preview

AI generated lesion overlay analysis

Mask Image

Overlay Image

AI Prediction Analysis

Prediction

Lesion Volume

Severity

Lesion Detected

320

Moderate MS

Report Summary

AI analysis detected 1 lesion(s) primarily located in the Periventricular region. The detected lesions show an estimated total lesion volume of 320 mm<sup>3</sup> with the largest lesion area measuring 80 pixels. Disease severity is currently classified as Moderate MS. Average lesion size is approximately 80 pixels, suggesting ongoing white matter abnormality patterns associated with Multiple Sclerosis. Further neurological and radiological evaluation is recommended for clinical confirmation and treatment planning.

AI MRI Lesion Detection System

Developed by Mousam Kashyap

Multiple Sclerosis MRI Analysis Report

Report Information

Patient ID

Uploaded Scan

MRI Slices Analyzed

1

AI Lesion Detection Summary

Lesion Slices Detected

Total Lesion Pixels

Estimated Lesion Volume (mm<sup>3</sup>)

Disease Severity

1

80

320

Moderate MS

MRI Lesion Detection Visualization

Input MRI

Detected Lesion Mask

Mask Overlay on MRI

III. Lesion Statistics

Total Lesions Detected

Average Lesion Size (pixels)

Largest Lesion Area (pixels)

Primary Brain Region

5

8000

11475

Periventricular

AI Assisted Diagnosis Tool

Research & Clinical Support

Generated using Deep Learning MRI Lesion Segmentation Model

Page 1 of 1

TURING BODY

Dashboard

Quick Actions

Recent Activity

Dr. Anvita

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

Total Patients

128

MRI Uploads

342

Weekly MRI Uploads

Mon

Tue

Wed

Thu

Fri

Sat

Positive Cases

89

Prediction Accuracy

95%

Quick Actions

Add Patient

Upload MRI

Generate Report

System Status

Recent Activity

MRI uploaded for Patient #1023

TURING BODY

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

AI MRI Upload System

Upload MRI scans and run AI-powered MS detection analysis.

MRI Scan Upload

Supported: JPG, PNG, DICOM

Drag & Drop MRI Scan

Upload brain MRI scans for AI-powered Multiple Sclerosis detection and analysis.

Browse MRI Files

Run AI Prediction

Patient Clinical Details

Patient ID

Checkup

Patient ID

Done

Weight (kg)

Height (cm)

Weight

Height

Clinical Notes

Doctor observations and clinical notes...

TURING BODY

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

Analytics & Reports

AI-powered MRI detection analytics and prediction insights.

Total MRI Scans

1,284

+12% this month

High-Risk Cases

342

Critical detections

Medium-Risk Cases

511

Moderate detections

Low-Risk Cases

431

Low-risk predictions

MRI Prediction Trends

Jan

Feb

Mar

Apr

May

Jun

AI Insights

High-risk lesions increased by 12%

Most affected age group: 30-45

AI confidence improved to 96%

MRI detection latency reduced by 18%

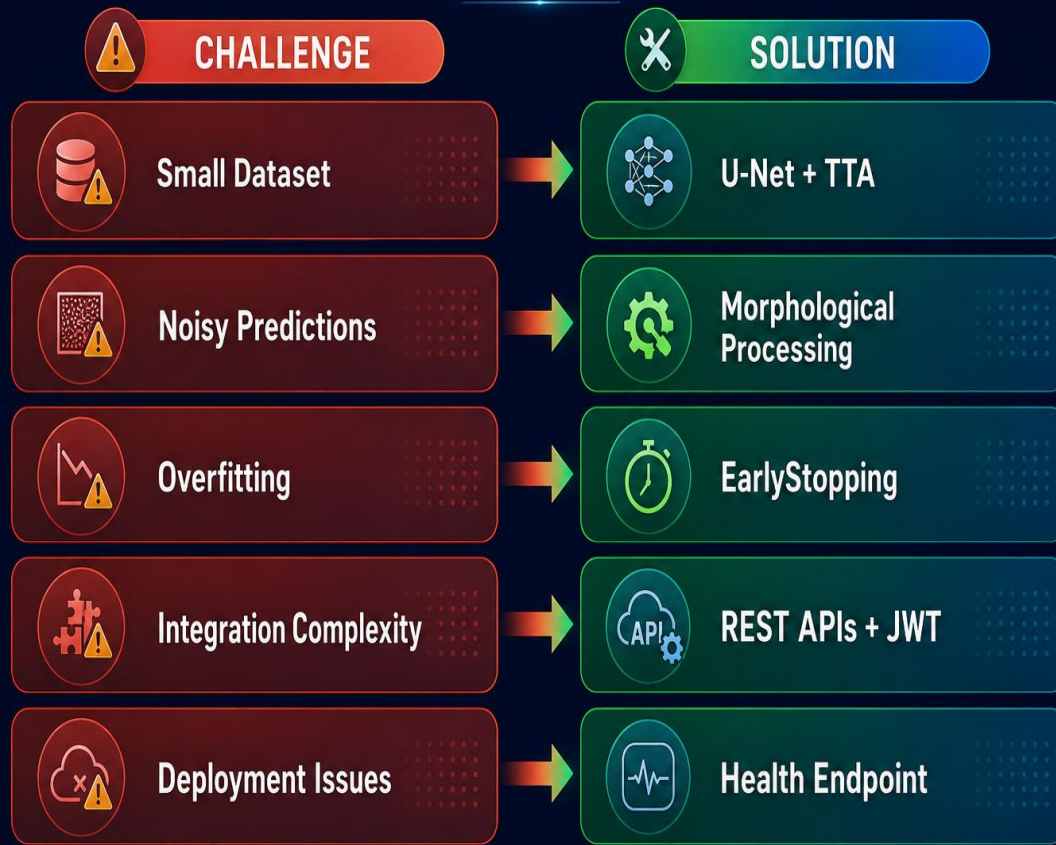
21-May-26

Project Presentation 2026

14

# CHALLENGES & SOLUTIONS

Overcoming Roadblocks. Building Reliable Solutions.



## Conclusion

- ❑ Turing Body successfully demonstrates the integration of Deep Learning, medical image processing, and full-stack web technologies for automated Multiple Sclerosis lesion detection and segmentation from MRI brain scans.
- ❑ The system utilizes a U-Net-based segmentation model to accurately identify lesion regions, analyze lesion characteristics, estimate disease severity, and generate visual PDF-based reports.
- ❑ By combining a Next.js frontend, Spring Boot backend, MongoDB database, and Python-based ML pipeline, the project provides a complete AI-powered healthcare platform that improves automation, medical visualization, and MRI analysis efficiency.
- ❑ The project highlights the practical application of Artificial Intelligence in medical imaging and showcases the potential of Deep Learning in assisting modern healthcare diagnosis workflows.

# Thank you